

Self-study assignment

For this week's self-study assignment we'll be applying some of the mathematical modelling techniques we've learnt in an example analysis. There are two options for the task, you only need to do **one**.

Option 1: 1918 influenza in Maryland

Look at `Maryland_incidence.csv` and `Maryland_population.csv`. These are data from the 1918 influenza pandemic in Maryland, United States. The first csv file shows the incidence (number of new cases) per day in 5 cities in Maryland. The second shows the population sizes for these cities.

Questions:

- 1) How would you model this influenza outbreak in Maryland? What SIR-type framework would be appropriate for modelling influenza?
- 2) What parameters do you need for your model? Can you find estimates for these from the literature?

Hint: How many people do you think were immune to the beginning?

- 3) Using these, generate some simulations for this scenario. How do these compare to the data?
- 4) What potential interventions could you model? There was
 - What would appropriate parameters be?
 - How many people do you think were immune at the beginning?
 - How do your simulations compare to the data?
 - What interventions could you model? The influenza vaccine was only developed in the 1930s but what if it had been available in the 1918 pandemic?

Option 2: Choose your own adventure

Choose a pathogen and a setting that interest you.

Questions:

- 1) What would the right model structure be for this pathogen? What do you need to add to a typical 'SIR'?
- 2) What parameters do you need to simulate the dynamics? Can you find estimates for these from the literature?

Hint: Think about population size, recovery time and delays

- 3) What (simple) interventions could you model?